From Gamma Synchrony to Informative Difference

A Shannon-Theoretic Model of Tonic Context and Phase-Wave Differentials

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Abstract

Gamma-band synchronization has long been investigated as a mechanism for feature binding, interareal communication, perceptual integration, and conscious access. That literature established an important operation: relative timing can coordinate distributed populations and selectively open a route from sender to receiver. Coordination, however, is not yet an account of differentiated content. Once a phase relation becomes a learned and repeatedly expected part of a local neural state, another unchanged cycle can be highly predictable and therefore carry little conditional self-information about what changed. This paper develops a complementary proposal from Self Aware Networks (SAN). A receiver-specific tonic sequence supplies context. Excitation, inhibition, altered timing, a missing expected event, a burst, a duration change, a spatial phase-gradient change, or a route change can create a less-probable physical departure. The departure becomes a candidate Phase-Wave Differential (PWD) when its dimensions are declared, its probability is evaluated against the receiver's learned reference model, and it produces a reproducible receiver-specific consequence.

The framework separates five quantities that are often compressed together: synchrony, sequence predictability, event surprisal, variability, and biological efficacy. Shannon self-information scores an event under a declared receiver, context, history, event alphabet, and probability model. Entropy rate scores the remaining uncertainty in a sequence after its history is known. Coefficient of variation summarizes relative dispersion in a selected positive linear variable; circular statistics are required for phase. A cross-validated likelihood ratio distinguishes a structured departure from both tonic continuation and rare noise. Receiver consequence and selective perturbation then test whether the event altered later recruitment, discrimination, memory, rendering, choice, or action. A preregistration-grade dLGN-to-V1 experiment is specified to compare the full model against rate, power, phase, phase-rate, conventional prediction-error, and matched-capacity alternatives. The proposal preserves the established value of gamma coordination while offering a falsifiable account of how coordinated neural activity can remain differentiated enough to carry changing content.

Keywords: gamma oscillation; synchrony; coherence; phase coding; entropy rate; Shannon information; inhibition; neural rendering; Phase-Wave Differential; Self Aware Networks

1. A coordinated brain still has to represent what changed

Imagine watching a steady red light turn green. The visual system cannot rebuild itself from zero at the moment of transition. Most of the body, room, retinal image, task, and ongoing neural state stay available as context. Yet some physical relation must distinguish the new moment from the old one. The brain therefore faces two simultaneous requirements:

1. distributed populations must remain coordinated enough to influence one another; and
2. their activity must remain differentiated enough to distinguish the present object, event, memory, or action from another.

Gamma synchronization became scientifically important because it offered a plausible mechanism for the first requirement. Stimulus-dependent synchrony was observed between visual cortical columns [4]. Coherent 40-Hz activity was proposed as a correlate of waking and dream cognition [5]. The temporal-correlation hypothesis developed synchrony as a binding operation [6]. Human studies then reported transient long-distance gamma synchronization during perception and strong uncoupling as a cognitive state changed [7], as well as early long-range gamma synchronization associated with conscious recognition in a visual task [8]. Later work showed that phase relations can regulate whether one population effectively communicates with another [11-16].

These findings solve part of the coordination problem. A recurrent rhythm can maintain a timing window. Phase alignment can make selected input arrive when a receiver is excitable. Differential coherence can select one sender while suppressing a competing sender. Distinct frequency channels can help separate feedforward and feedback influence. But those operations leave a further question:

> *Once a route is coordinated, which physical event says that the present content differs from the expected content?*

A clock can keep a route open without specifying every message sent through it. In the same way, a highly regular synchronized relation can be functionally essential while its next unchanged cycle is unsurprising. The next cycle may still identify a state, route, or carrier. It simply provides little new evidence that the current relation changed.

The proposal developed here begins with that distinction. A maintained oscillatory relation supplies context. A measurable excitation- or inhibition-driven departure supplies a candidate change. A receiver-specific probability model measures how unusual that change is. A receiver-specific consequence tests whether the difference mattered biologically. Only after that operation is clear will the paper name the resulting event a **Phase-Wave Differential**.

2. What the gamma literature established

2.1 Synchrony can integrate and route

The history of gamma research is not one claim but a progression of operations. Gray and colleagues showed that visual cortical columns could synchronize in a stimulus-dependent way [4]. Crick and Koch proposed coherent semi-synchronous oscillations in the 40-70 Hz range as part of a neurobiological account of visual awareness [3]. Llinás and Ribary reported coherent 40-Hz magnetic activity in waking and REM sleep and proposed that specific thalamocortical loops contribute content while nonspecific loops contribute temporal binding [5]. Joliot, Ribary, and Llinas further reported that human activity near 40 Hz covaried with cognitive temporal binding [34]. Singer and Gray developed the temporal-correlation hypothesis as an account of feature integration [6].

Rodriguez and colleagues then reported long-distance 30-80 Hz synchronization during face perception, followed by strong desynchronization as processing moved toward a motor response [7]. Their interpretation already contained both coordination and active uncoupling. Melloni and colleagues found local gamma increases for visible and invisible words but an early transient of long-distance gamma phase synchrony specifically for perceived words [8]. Doesburg and colleagues reported gamma-synchronous assemblies that formed and dissolved with perceptual switching during binocular rivalry [9]. These were substantive positive results, but later dissociations narrowed their interpretation. Aru and colleagues found local category-specific gamma responses that did not track conscious perception [35]. Pitts and colleagues independently varied awareness and task relevance; induced gamma was absent for task-irrelevant stimuli regardless of awareness and robust for task-relevant stimuli, supporting a post-perceptual or task-related interpretation rather than a unique awareness marker [36]. In a different memory preparation, Solomon and colleagues found widespread theta synchrony alongside high-frequency desynchronization during enhanced cognition [19]. Gamma therefore remained a serious coordination and correlation program without becoming a sufficient specification of conscious content or reducing cognition to maximum high-frequency synchrony.

Communication-through-coherence supplied a mechanistic routing account: coherent sender activity arriving at a favorable receiver phase can exert greater influence [11]. Womelsdorf and colleagues showed that neuronal interaction depended on phase relation and that the interaction-supporting relation preceded effective interaction [12]. Grothe and colleagues found that one V4 population could show high gamma coherence with the task-relevant V1 input and low coherence with a competing input, even after equalizing V1 power [13]. Bastos and colleagues found distinct frequency channels for feedforward and feedback influences across primate visual areas [14]. Zandvakili and Kohn found that brief V1 coordination was followed by spiking specifically in the downstream V2 input layer, showing that coordination can be anatomically selective rather than globally propagated [15]. Palmigiano and colleagues modeled transient synchrony as a flexible routing mechanism [16].

Together, this literature supports three established conclusions:

- gamma-range coordination can accompany perceptual and cognitive operations;
- relative phase can alter effective sender-receiver communication; and
- useful coordination can be transient, selective, and followed by uncoupling or reconfiguration.

2.2 Gamma is not one measurement

The word *gamma* can conceal different physical signals. Ray and Maunsell separated narrowband gamma rhythm from broadband high-gamma activity in macaque V1 and found that high-gamma power tracked local spiking while narrowband gamma could move in the opposite direction [21]. Yuval-Greenberg and colleagues showed that a commonly reported transient scalp-EEG induced-gamma response could be produced by microsaccadic spike potentials [20]. These findings do not negate neuronal gamma rhythms. They require every gamma claim to state its sensor, bandwidth, reference, artifact controls, and biological scale.

The present framework therefore avoids treating EEG gamma power, LFP phase, spike-field coupling, interareal coherence, dendritic voltage, and spike timing as interchangeable. They are different measurements that may enter a joined mechanism only after their relation has been tested.

2.3 Integration and differentiation were already recognized as a paired problem

Tononi and Edelman argued that a neural process relevant to consciousness must combine functional integration with differentiation [10]. That earlier conceptual boundary matters. The contribution developed here is not the general observation that consciousness requires both unity and variety. It is an event-level oscillatory mechanism that asks how a maintained receiver-specific relation, an uncommon typed departure, and a downstream physical consequence can instantiate part of that pairing.

3. The missing information question

Suppose two populations have learned a stable phase relation. That relation may identify a route, maintain readiness, or increase effective coupling. Once it is expected, however, its continuation is predictable from recent history. By contrast, a phase advance, delayed event, missing expected event, added burst, changed duration, or shifted route may be less probable under the same history.

Shannon's probability-information relation supplies a precise language for the contrast [1,2]. It does not declare which neural event is meaningful. It requires the experimenter to identify a receiver, a context, an event alphabet, and a probability distribution. Under that declared model, less-probable events have greater self-information.

Neuroscience already used this framework before SAN. Strong and colleagues measured entropy and information in neural spike trains [18]. Montemurro and colleagues found that spike phase relative to a low-frequency LFP carried additional stimulus information beyond spike counts in their macaque V1 preparation [17]. The novel burden is therefore more specific than applying Shannon information to a neural signal. The missing joined operation is:

```
expected receiver-specific oscillatory context
-> typed excitatory or inhibitory departure
-> conditional event rarity
-> structured rather than random difference
-> receiver-specific physical consequence
-> changed downstream content or action
```

This sequence separates a carrier from a change, a rare event from random noise, and decodability from causal efficacy.

4. The operation in plain language

4.1 The maintained relation supplies context

A receiver does not encounter each input as if nothing happened before. Its membrane state, inhibition, morphology, receptor distribution, recent firing, surrounding field activity, and network history affect what an arriving signal can do. Recurrent coordination can make part of that state stable enough to predict what normally comes next. This paper calls that functional reference the **tonic relation**. Tonic is a role, not a permanent frequency band. A repeated gamma relation can be tonic for one process, while a slower event can be phasic if it is the unexpected change.

4.2 Excitation and inhibition can both create difference

An excitatory event can advance timing, add a burst, recruit more cells, lengthen activity, or increase a receiver's drive. Inhibition can delay an expected event, suppress a route, sharpen a population pattern, or create an informative omission. Inhibitory circuits can synchronize inhibition [22], divide or subtract components of cortical responses [23], and organize distinct beta- and gamma-band dynamics [25,26]. Timing and post-induction activity also constrain whether a plastic change persists in a spike-timing-dependent preparation [24]. An omission is therefore not mere nothingness when a receiver has learned that an event should occur at that time.

4.3 Repetition and departure answer different questions

The tonic relation answers: *What state and route are currently maintained?* The departure answers: *What changed relative to that maintained state?* A synchronized event can play either role. A stable nonzero phase offset can also be highly coordinated. Conversely, broad random phase dispersion can be unusual without carrying useful content. The relevant contrast is expected versus unexpected under a declared model, followed by structured versus random and effective versus ineffective.

4.4 Naming the joined event

SAN names the joined, receiver-relative event a **Phase-Wave Differential (PWD)**. The name does not mean a generic phase code or every traveling wave. It denotes a physical difference evaluated against a receiver's current reference relation, described across the measured event dimensions, and followed through its consequence. The PWD may include phase, frequency, amplitude, duration, transmitted quantity, spatial phase gradient, route, or a missing expected event. Its mechanistic status depends on what it changes in the receiver and what happens next.

5. Formal model

5.1 Receiver, context, history, and event alphabet

Let r denote the receiver, c_t the declared context, h_t the measured history available before time t, and $\mathbf{x}_t$ the event features in a preregistered alphabet. The reference model is

$$p_r(\mathbf{x}_t \mid h_t, c_t). \tag{1}$$

The context may include task epoch, arousal, movement, pupil diameter, recent stimulus sequence, neuromodulatory proxy, and the selected network's tonic phase relation. Training data used to fit Equation (1) must be separated from held-out evaluation data. For continuous variables, the analysis must use preregistered bins or a fixed reference measure and units; an isolated probability density is not an invariant number of bits under arbitrary unit changes.

5.2 Conditional self-information

For an observed event, define receiver-conditioned self-information

$$\mathcal{I}_r(\mathbf{x}_t \mid h_t, c_t) = -\log_2 p_r(\mathbf{x}_t \mid h_t, c_t). \tag{2}$$

An expected continuation has high conditional probability and low surprise about change. An advance, delay, omission, burst, duration shift, amplitude shift, phase-gradient change, or route switch can have lower conditional probability and greater self-information. Equation (2) is a probabilistic score. Biological meaning enters only after the event's structure and consequence are tested.

5.3 Sequence predictability and entropy rate

The user-facing intuition concerns repetition across time, not only one isolated probability. For a finite history length m, define the conditional entropy rate estimate

$$h_r^{(m)} = H(\mathbf{X}_t \mid \mathbf{X}_{t-m:t-1}, C_t, R = r). \tag{3}$$

Under appropriate stationarity and convergence assumptions, the process entropy rate is

$$h_r = \lim_{m \to \infty} h_r^{(m)}. \tag{4}$$

If a phase-locked sequence repeats deterministically and its state is known, the uncertainty about its next continuation approaches zero under that model. The sequence can still identify its onset, state, route, or relation to another variable. The low-entropy-rate statement concerns uncertainty about continuation after the relevant history is known; it is not an absolute declaration that synchrony has no function or no information under every question.

5.4 A minimal typed departure vector

Let $\bar{\mathbf{x}}_{r,c,h}$ denote the receiver's expected relation. A paper-specific typed departure vector is

$$\mathbf{d}_{r,t} = \begin{bmatrix} \mathrm{wrap}(\phi_t - \bar{\phi}_{r,c,h}) \\ f_t - \bar{f}_{r,c,h} \\ A_t - \bar{A}_{r,c,h} \\ \tau_t - \bar{\tau}_{r,c,h} \\ q_t - \bar{q}_{r,c,h} \\ \nabla\phi_t - \overline{\nabla\phi}_{r,c,h} \\ o_t \\ \Delta\mathbf{z}_{r,t} \end{bmatrix}, \tag{5}$$

where o_t marks an expected-event omission and $\Delta\mathbf{z}_{r,t}$ contains preregistered route or receiver-state variables. Not every experiment requires every component. The complete numerical encoding contract belongs to the companion *Phase-Wave Differential Calculus* paper; the minimal vector here prevents a post hoc decision to call every unusual signal a PWD.

The event record is

$$\mathrm{PWD}_{r,t} = (\mathbf{d}_{r,t}, \mathcal{I}_r(\mathbf{x}_t \mid h_t, c_t), r, h_t, c_t). \tag{6}$$

Equation (6) defines a candidate measurement object. Mechanistic status additionally requires a receiver consequence.

5.5 Structured departure versus rare noise

Rarity alone cannot distinguish signal, artifact, pathology, or noise. Fit three matched-capacity models on training data: M_T for tonic continuation, M_D for a structured departure class, and M_N for a rate- and power-matched noise class. For a held-out event, define

$$\Lambda_{D:T}(t) = \log_2 \frac{p(\mathbf{x}_t \mid M_D, h_t, c_t, r)}{p(\mathbf{x}_t \mid M_T, h_t, c_t, r)}, \qquad \Lambda_{D:N}(t) = \log_2 \frac{p(\mathbf{x}_t \mid M_D, h_t, c_t, r)}{p(\mathbf{x}_t \mid M_N, h_t, c_t, r)}. \tag{7}$$

A candidate structured departure should outperform both unchanged continuation and matched random perturbation on held-out data. This test does not require every meaningful event to be common enough to form a pre-existing class; it requires the event-generating rule and evaluation to be frozen before the confirmatory trials are opened.

5.6 How coefficient of variation connects without becoming a synonym

For a positive linear feature X whose reference mean remains safely away from zero, define the event-level relative residual

$$\rho_{X,t} = \frac{X_t - \mu_{X,r,c,h}}{|\mu_{X,r,c,h}|}. \tag{8}$$

Across a reference window whose mean is μ_X, the root-mean-square relative residual is

$$\sqrt{\mathbb{E}[\rho_X^2]} = \frac{\sigma_X}{|\mu_X|} = CV_X. \tag{9}$$

Equation (9) makes the connection precise. Event-level relative departures form the distribution whose RMS is summarized by coefficient of variation. CV is a window statistic for one selected positive linear variable; it is not the event, its Shannon information, or its biological effect. It becomes unstable near a zero mean and is unsuitable for circular phase.

For phase samples θ_k, define the mean resultant and circular variance

$$Re^{i\bar{\theta}} = \frac{1}{n}\sum_{k=1}^{n} e^{i\theta_k}, \qquad V_\theta = 1 - R. \tag{10}$$

High R means a concentrated phase relation, not necessarily zero-lag synchrony. A reproducible nonzero offset can be strongly phase locked. Low R indicates broad dispersion but does not by itself say whether the dispersion is meaningful or random.

5.7 Receiver consequence

Let $\mathbf{s}_{r,t}$ be the receiver's state and F_r its local transformation. Define the receiver-specific consequence of the departure as

$$\Delta\mathbf{y}_{r,t+\delta} = F_r(\mathbf{s}_{r,t}, \mathbf{d}_{r,t}) - F_r(\mathbf{s}_{r,t}, \mathbf{0}). \tag{11}$$

The measured consequence may be a membrane-voltage change, dendritic event, spike-time shift, release-probability change, altered recruitment, route suppression, phase reset, plasticity, or a later behavioral effect. The same physical event can affect receivers differently because their morphology, receptors, recent history, inhibition, and tonic state differ.

Incremental predictive value can be summarized by

$$I(\mathbf{D}; \mathbf{Y} \mid \mathbf{C}, \mathbf{H}, \mathbf{B}), \tag{12}$$

where $\mathbf{B}$ contains rate, power, movement, arousal, and common-input baselines. Equation (12) is not yet causal. Causal status requires selective perturbation and matched rescue.

6. How this operation differs from neighboring terms

The vocabulary earns its place only if it performs work that existing terms do not already perform.

Account	Measured object	What it establishes	What the PWD model adds
Gamma synchrony	phase relation among signals	coordination or integration	departure from a learned relation, event rarity, receiver consequence
Communication through coherence	sender-receiver phase alignment	route effectiveness in a timing window	content-bearing change within or against the window
Phase-of-firing code	spike timing relative to oscillatory phase	decodable stimulus information	typed nonphase dimensions, omission events, receiver history, downstream efficacy
Traveling-wave account	propagation of phase across space	movement and geometry of population activity	receiver-relative deviation, conditional probability, and consequence
Prediction error	mismatch between expected and observed variables	model-relative discrepancy	a physically typed neural event, including inhibition and route state, measured at a declared receiver
CV or dispersion	variability of one selected feature	spread relative to a scale	event identity, Shannon score, circular phase handling, route, and effect
Neural complexity or integrated information	integration and differentiation at system level	global or subset-level organizational property	event-scale oscillatory implementation connecting tonic context to specific changing content

The point of the PWD term is therefore not to rename phase coding or traveling waves. It names a larger operation whose parts can be measured separately and whose added value can fail.

7. The dated SAN development

The present equations are a 2026 formalization. The mechanism developed through separately dated public stages in 2022.

7.1 June 8, 2022: phasic and inhibitory departures against tonic activity

In `a0027z`, I described the brain as rendering phasic burst-firing patterns as deviations from tonic firing and stated that inhibition also contributes to the rendering [29]. This source fixes the context-versus-departure operation. It precedes the explicit rarity equation and the named PWD.

7.2 June 11, 2022: rarity and information value

In `a0245z`, I described tonic firing as a canvas and phasic firing as changing ink. I explicitly connected the relative rarity of phasic events to greater information value while retaining tonic activity as the context that makes the departure interpretable [30].

7.3 June 27, 2022: expected periodicity and physical surprise

In `a0206z`, I treated tonic or periodic activity as an expected interval and described both an added event and an inhibited expected event as a physical departure [31]. This stage makes omission part of the candidate event alphabet.

7.4 July 9, 2022: receiver projection and developing NAPOT

In `a0149z`, I joined frequency and duration variation, phase differences as information states, and projection from one neural array to a receiving array in the developing Neural Array Projection Oscillation Tomography (NAPOT) account [32]. The event's significance depends on what the receiving array detects and transforms.

7.5 August 23, 2022: the named operator

In `a0306z`, the exact term **Phase Wave Differential** is publicly fixed and joined to tonic, phasic, high-phasic, and inhibited phase patterns; dendritic detection; array-to-array re-expression; and multimodal object properties [33]. The June and July sources are verified mechanism ancestors. August 23 is the currently verified public fixation of the name.

The source sequence establishes the following historical signature without assigning later notation to earlier prose:

```
tonic context
-> phasic and inhibitory departure
-> rarity as information value
-> periodic expectation and omission
-> phase, frequency, duration, and receiver projection
-> named PWD in a rendering-to-action architecture
```

8. From informative difference to neural rendering and action

SAN uses this event inside a no-inner-observer architecture. The network does not send a completed semantic object to a hidden viewer. Each receiving cell or array is physically altered according to its own state, then re-expresses part of the transformed relation downstream. Recurrent exchange can stabilize some relations, inhibit others, and form temporary assemblies around the current sensory, memory, or action problem.

The mechanism separates several jobs:

- **coherence** makes selected sender-receiver interaction effective;
- **structured separation** preserves distinct routes or candidates;
- **excitation and inhibition** create typed differences relative to expectation;
- **conditional surprisal** measures how uncommon the event is under the declared history;
- **receiver tuning** determines the event's local transformation;
- **recurrent propagation** amplifies, suppresses, stores, or redirects the consequence.

The resulting chain is

```
broad tonic coordination
-> phasic excitatory or inhibitory departure
-> phase-compatible summation or structured separation
-> changed receiver state
-> changed population recruitment
-> neural rendering memory choice or motor consequence
```

This chain also clarifies the relation between coincidence and action. Phase-compatible inputs that arrive within a receiver's integration window can sum toward threshold. Downstream action potentials then recruit motor pools; neuromuscular transmission triggers calcium-dependent, ATP-powered muscle contraction. Neural timing organizes recruitment, while metabolic chemistry supplies muscular energy.

9. Prospective confirmatory dLGN-to-V1 test

9.1 Experimental question

Does a receiver-conditioned model of typed departures predict and causally control V1 responses and visual detection better than rate, power, phase, phase-rate, conventional prediction-error, and matched-capacity multivariate alternatives?

The initial preparation tests the component mechanism rather than consciousness as a whole. The dorsal lateral geniculate nucleus (dLGN) to V1 route is selected because simultaneous awake-mouse recordings have already shown high-gamma phase synchronization and precise spike synchrony across this pathway [27]. Closed-loop phase-targeted optogenetic modulation of excitatory and inhibitory cortical populations has also been demonstrated in awake mice [28].

9.2 Animals and independent unit

The confirmatory cohort will contain **20 quality-control-passing adult mice**, with up to 24 enrolled to account for surgical or recording attrition. Sex will be balanced where colony yield permits and included as a prespecified covariate. The animal, not the trial, is the independent unit. Four separate pilot animals may be used to calibrate event timing, closed-loop latency, and the reference distribution; they will never enter confirmatory estimates.

The confirmatory design targets eight valid sessions and 280 accepted trials per session in each animal: 40 trials in each of seven conditions, or 2,240 accepted trials per animal. A deterministic Monte Carlo calibration estimates 0.92479 two-sided power for 20 passing animals at a standardized paired animal-level effect of $d_z = 0.8$, using 200,000 simulations and seed 20260809. This is a prospective design calibration, not an experimental result; its complete machine-readable receipt is provided as Supplementary File S1. A hierarchical simulation using blinded pilot variance and event prevalence will be frozen before confirmatory condition labels are opened. If that blinded simulation requires more than 20 passing animals, enrollment can increase, up to the stated maximum of 24, only through a prospectively timestamped protocol amendment made before analysis.

9.3 Behavior and stimuli

Head-fixed mice will report a visual sequence change by releasing a wheel hold or licking within a fixed response window. Each session begins with a high-contrast grating whose 2-Hz contrast-reversal sequence establishes a stable dLGN-to-V1 relation. The 2-Hz stimulus sequence is the learned visual schedule; it is not the neural gamma frequency. Each animal's neural gamma band will be localized independently within 30-90 Hz.

Condition	Visual event	Closed-loop intervention	Preserved comparison
C1 expected continuation	scheduled reversal at 500 ms	none or sham	tonic reference
C2 advance	reversal 125 ms early	none	timing departure at matched event count
C3 delay	reversal 125 ms late	none	timing departure at matched event count
C4 omission	scheduled reversal withheld	none	missing expected event; analyzed separately
C5 additional event	extra reversal halfway through the interval	none	added event; analyzed separately
C6 structured neural retiming	expected visual continuation	fixed pulse train at the preregistered effective gamma phase	reproducible nonzero phase relation at matched pulse count and light energy
C7 matched random retiming	expected visual continuation	identical pulse train at randomized phase	temporal-structure null at matched pulse count, light energy, and gross rate target

Conditions will be balanced, randomly interleaved, and generated from a preregistered seed family. Stimulus identity, contrast, luminance, and gross event count will be matched wherever the hypothesis permits. Omission and additional-event trials necessarily differ in event count and therefore will not be advertised as rate-matched evidence; those contrasts will be analyzed separately from the visual-retiming and neural-retiming contrasts.

9.4 Recording and state controls

Simultaneous Neuropixels recordings will sample dLGN sender activity and V1 receiver layers. LFP, single-unit and multiunit spikes, spike-field locking, current-source-density estimates, wheel movement, pupil diameter, eye position, whisking, respiration, and licking will be acquired on one clock. The stimulation trigger, command, photodiode confirmation, and measured light output will be recorded on that same acquisition clock. An independent localizer will identify each animal's visually driven narrowband oscillatory peak within 30-90 Hz; the confirmatory band cannot be moved after outcome labels are inspected.

Preprocessing will include probe drift correction, spike-quality thresholds, line-noise handling, microsaccade or eye-movement covariates, movement exclusion windows, and artifact rejection fixed before model comparison. Narrowband gamma and broadband high-gamma will be modeled separately.

The primary receiver outcome is the held-out 5-ms-bin spiking state of histologically registered V1 layer-4 regular-spiking units during the 0-150 ms interval after the scheduled or observed visual transition. Log loss will be averaged within unit, session, and animal before group inference. Laminar current-source density, spike-field coupling, population decoding, and behavioral detection are secondary outcomes.

9.5 Event alphabet and reference model

The event alphabet will include wrapped dLGN-V1 phase residual, inter-event interval, instantaneous frequency, narrowband amplitude, broadband high-gamma power, spike count, burst duration, omission indicator, laminar receiver location, and pre-event receiver state. The tonic model will be trained only on expected-continuation trials and pre-event history. Bins, kernel widths, phase estimator, history length, and missing-event window will be frozen using pilot data.

9.6 Model comparison

All models will use the same outer animal/session folds, behavioral-state covariates, and nested training procedure. Hyperparameters will be selected only inside training folds. Model capacity will be compared at shared effective degrees of freedom under the same regularization family; if exact matching is impossible, the complete shared-capacity curve will be reported rather than selecting a single favorable operating point:

1. rate and spike-count model;
2. narrowband power model;
3. phase-only model;
4. joint phase-rate model;
5. conventional task prediction-error model;
6. matched-capacity multivariate neural-state model without PWD structure;
7. surprisal-only model;
8. PWD model without receiver consequence; and
9. full PWD model with receiver state and downstream consequence.

The primary predictive endpoint is animal-level held-out log-loss improvement in V1 receiver-event prediction over the strongest baseline:

$$\Delta L_{PWD} = L_{test}(M_{best\ baseline}) - L_{test}(M_{PWD}). \tag{13}$$

The predictive gate passes only if the median ΔL_{PWD} is positive across animals, the hierarchical 95% interval excludes zero, the result replicates in a temporally held-out session block, calibration error remains within its preregistered tolerance, and no surrogate null reproduces the result. Surrogates include condition-label permutation within arousal and movement strata, circular time shifts beyond the receiver window, phase randomization with the power spectrum preserved, rate- and power-matched temporal jitter, within-layer receiver-identity permutation, history truncation, a future-variable leakage test, an eye-movement-only model, and a broadband-spike-transient model.

9.7 Closed-loop perturbation and rescue

After the predictive gate is frozen, closed-loop stimulation will retime the selected V1 excitatory or inhibitory population at a preregistered phase while holding pulse count and light energy constant. The system must demonstrate end-to-end timing jitter below 2 ms in calibration; otherwise the phase-specific causal claim is suspended and only the observational analysis proceeds.

For an expected event, the perturbation should selectively disrupt the predicted receiver relation. For an omission event, phase-matched stimulation will attempt to restore the expected receiver pattern without restoring the visual stimulus. For a retimed event, matched rescue will return the stimulation to the predicted effective phase. Causal effect is summarized as

$$\mathrm{ACE}_{PWD} = \mathbb{E}[Y \mid do(\mathbf{D} = \mathbf{d}_1)] - \mathbb{E}[Y \mid do(\mathbf{D} = \mathbf{d}_0)]. \tag{14}$$

The causal gate passes only when perturbation produces the preregistered receiver-specific failure, phase-matched rescue reverses it, and open-loop replay, random-phase stimulation, sham light, gross rate, total power, movement, and arousal fail to explain the pattern.

9.8 Statistical gatekeeping and exclusions

The predictive test is primary and is tested first at two-sided alpha 0.05. The causal test is interpreted confirmatorily only after the predictive gate passes and is then tested at two-sided alpha 0.05. All remaining contrasts are secondary or exploratory and receive multiplicity-adjusted intervals. This sequential rule avoids treating many exploratory contrasts as independent confirmations. Animal exclusions are limited to preregistered surgical, histological, opsin-expression, behavioral, and signal-quality criteria. Every enrolled animal, excluded animal, session attempt, accepted trial, model degree of freedom, protocol amendment, and analysis change will be reported.

9.9 Human consciousness extension

A later human extension can use masking or binocular rivalry with simultaneous eye tracking and MEG, high-density EEG, or intracranial recordings. It should compare reported or no-report perceptual transitions against expected continuations while separating narrowband neuronal gamma from microsaccadic and broadband artifacts. The consciousness claim advances only if receiver-conditioned departures predict content-specific transitions beyond report, movement, power, rate, and stimulus differences.

10. Predictions and falsifiers

The theory makes the following predictions.

1. **Sequence prediction:** an established tonic relation will have lower held-out conditional entropy about continuation than structured departure conditions.
2. **Rarity without equivalence:** self-information and biological consequence will be correlated only for a subset of events; matched random jitter may be rare but ineffective.
3. **Receiver dependence:** the same departure will have different consequences across receiver layers, cell types, and pre-event states.
4. **Omission as event:** a learned missing event will be decodable in receiver state and will alter later processing when the route is behaviorally relevant.
5. **Added model value:** the full PWD model will improve held-out prediction beyond rate, power, phase, phase-rate, task prediction error, and matched multivariate state.
6. **Selective perturbation:** phase-specific intervention will disrupt the predicted receiver and matched rescue will restore it better than open-loop or random-phase controls.
7. **Structured separation:** a reproducible nonzero phase relation can carry stable route identity even when zero-lag synchrony is absent.

The stronger PWD account is weakened or rejected if any of the following persists across adequate data and controls:

- the full model fails to outperform matched baselines on held-out animals or sessions;
- rarity adds nothing once rate, power, history, movement, and common input are controlled;
- structured departures are indistinguishable from matched random jitter;
- receiver state does not modify event consequence;
- perturbation effects follow light energy or firing rate rather than predicted phase and route;
- rescue does not restore the receiver-specific outcome; or
- the putative gamma effect is explained by broadband spiking, eye movement, filtering, or another artifact.

11. Implications for consciousness science

Gamma synchronization may be a real part of conscious neural dynamics without being the complete physical content of a conscious moment. A coordinated relation can supply unity, route availability, or temporal context. Differentiated content requires relations that vary in ways the participating receivers can detect and use.

This proposal therefore reframes a long-standing difficulty. The failure of gamma synchrony alone to identify the content of consciousness does not make synchronization irrelevant. It suggests that coordination and distinction are complementary operations. A learned synchronized relation can be the low-surprise canvas. Excitatory and inhibitory departures can be the changing differences. The receiving network, rather than a hidden inner observer, is altered by those differences and carries their consequences forward.

This view also predicts that excessive global coherence can reduce differentiated capacity even while increasing coordination. Conversely, broad incoherence can destroy usable routing. The candidate conscious regime is not maximum synchrony or maximum disorder. It is coordinated context with structured, receiver-effective differentiation. That proposition connects gamma research to information theory without reducing consciousness to a single frequency or a single scalar.

12. Interpretation boundaries

Several distinctions keep the proposal testable.

First, **low conditional surprise is question-specific**. A repeated cycle may be unsurprising about change while remaining informative about state, identity, timing, or route.

Second, **high surprise is not meaning**. Artifact, pathology, and random noise may be rare. The structured-departure and receiver-consequence gates are essential.

Third, **phase locking is not restricted to zero lag**. A stable nonzero phase offset can be highly coordinated. Circular dispersion and mean phase must be reported together.

Fourth, **CV is a local distribution summary**. It applies only to a declared linear feature with a safe mean. Phase requires circular statistics, and multivariate departures require an explicit joint model.

Fifth, **tonic and phasic are functional roles**. They cannot be permanently assigned to slow and fast bands.

Sixth, **measurement scales remain distinct**. EEG, MEG, LFP, spikes, dendritic voltage, synaptic release, and local electric fields cannot substitute for one another without a validated mapping.

Finally, **component success is not whole-theory proof**. Predicting and controlling a receiver-relative departure would establish a candidate neural mechanism. A content-specific consciousness claim additionally requires perceptual endpoints and artifact-resistant controls. The larger SAN architecture remains responsible for explaining how such events participate in a distributed self-aware rendering.

13. Conclusion

Gamma synchronization helped neuroscience explain how distributed populations can coordinate and selectively communicate. Shannon information clarifies why coordination is not identical to changing content. Once a receiver-specific oscillatory sequence becomes expected, its unchanged continuation can have low entropy rate and low conditional surprise about change. A less-common excitation- or inhibition-driven event can carry greater self-information, but it becomes a mechanistic signal only when it is structured, receiver-effective, and causally connected to what happens next.

The Phase-Wave Differential names that joined event: a typed physical departure measured against a receiver's tonic distribution and followed through its consequence. This paper turns the proposal into a sequence model, a structured-versus-noise discriminator, a precise CV bridge, and a preregisterable perturbation-and-rescue experiment. The central empirical demand is simple: the full model must predict and control receiver consequences better than rate, power, phase, phase-rate, ordinary prediction error, and matched multivariate alternatives. If it fails, the added mechanism does not earn its name. If it succeeds, gamma coordination and informative difference become complementary parts of a more complete account of neural content.

Data, code, and preregistration availability

This article presents a theory and prospective experimental protocol. It reports no new animal or human data. The release package includes the protocol, frozen model-gate specification, machine-readable power calibration, source ledger, claim-source matrix, and validation receipts. The confirmatory protocol should be publicly preregistered before data collection begins; the present manuscript does not represent that registration as already completed.

Supplementary materials

- **S1:** deterministic animal-level power calibration and simulation receipt.
- **S2:** prospective confirmatory dLGN-to-V1 protocol.
- **S3:** machine-readable model, predictive-gate, surrogate-null, and causal-gate specification.
- **S4:** claim-source matrix and source-custody ledger.

Research chronology

The Self Aware Networks concepts developed here were created and documented by Micah Blumberg in a research program beginning before the named PWD formulation. The public source record cited below fixes the tonic-context, rare-departure, expected-omission, receiver-projection, and named-PWD stages separately between June and August 2022. Current equations and experimental controls are later formalizations and are dated as such.

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